Validation against published sources

mortgagemath 0.6.1 · rendered 2026-05-06

Every fixture in the table below reproduces **every** value its source publishes, to the cent. The library follows a strict policy: if any published value diverges, the fixture stays out. The six parameter columns show the exact `LoanParams` settings required to match each source — the same six knobs documented in the *at-a-glance* vignette.

Validated fixtures

Source	Loan	Day	Bal	Rnd	Cmp	Freq	ARM
Arcones SOA FM §4.1 Ex 4	$20k / 8% / 12yr	30/360	RE	HU	A	A	–
Broverman MIC 2.7(a)	$12k / 12% / 3yr	30/360	RE	HU	M	M	–
Broverman MIC 2.7(b)	$12k / 15% / 4yr	30/360	RE	HU	M	M	–
CFPB H-25(B)	$162k / 3.875% / 30yr	30/360	RE	HU	M	M	–
CFPB IO Sample	$211k / 4% / 30yr	30/360	RE	HU	M	M	–
eCampus §4.3 Pearline	$10k / 10% / 4yr	30/360	RE	HU	A	A	–
eCampus §4.3 Erika	$32.6k / 4.83% / 9yr	30/360	RE	UP	M	M	–
eCampus §4.3 Johnetta	$20.2k / 3.53% / 8yr	30/360	RE	HU	M	M	–
eCampus §4.4.1 T1	$297.5k / 3.8% / 3yr, 20yr amort	30/360	RE	HU	j_2	Q	–
eCampus §4.4.1 renew	$265.8k / 2.5% / 17yr	30/360	RE	HU	j_2	Q	–
FNMA §1103	$25M / 5.5% / 10yr, 30yr amort	A/360	RE	HU	M	M	–
FHLBB 1935 Plan A	$3k / 6% / 139mo	30/360	CP	HU	M	M	–
Geltner Ch 20	$1M / 12% / 30yr	30/360	CP	HU	M	M	–
Goldstein §10.3	$563 / 12% / 5mo	30/360	CP	HU	M	M	–
JHF Flat 35	$20M / 1.5% / 30yr	30/360	RE	HU	M	M	–
Las Positas §8.05 Ex 1	$15k / 9% / 5yr	30/360	RE	UP	M	M	–
Las Positas §8.05 Ex 3	$18k / 2% / 5yr	30/360	RE	UP	M	M	–
MoneyVox FR	$10k / 5% / 1yr	30/360	RE	HU	M	M	–
MS State P3920	$100k / 7% / 30yr	30/360	RE	HU	M	M	–
Olivier Chans T1	$350.1k / 4.9% / 3yr, 20yr amort	30/360	RE	HU	j_2	M	–
Olivier Chans renew	$316.6k / 5.85% / 17yr	30/360	RE	HU	j_2	M	–
OpenStax §6.12.110	$132.7k / 4.8% / 30yr	30/360	RE	UP	M	M	–
OpenStax AK 6.100.1	$18.3k / 6.75% / 4yr	30/360	RE	UP	M	M	–
OpenStax AK 6.100.2	$41.6k / 3.9% / 6yr	30/360	RE	UP	M	M	–
OpenStax AK 6.110	$153.9k / 4.21% / 20yr	30/360	RE	UP	M	M	–
OpenStax AK 6.114	$159.2k / 5.75% / 30yr	30/360	RE	UP	M	M	–
OpenStax AK 6.36	$23.7k / 4.76% / 5yr	30/360	RE	UP	M	M	–
OpenStax AK 6.78.1	$17.9k / 7.5% / 10yr	30/360	RE	UP	M	M	–
OpenStax AK 6.78.2	$33.8k / 4.3% / 20yr	30/360	RE	UP	M	M	–
OpenStax §6.8 car	$28.5k / 3.99% / 5yr	30/360	RE	UP	M	M	–
OpenStax §6.8 home	$136.7k / 5.75% / 15yr	30/360	RE	UP	M	M	–
ProEducate cap	$65k / 10% / 30yr	30/360	RE	HU	M	M	1, cap
RBC Acc Bi-Weekly	$350k / 5% / 25yr	30/360	RE	HU	j_2	BW	–
Reg Z H-14	$10k / 17.41% / 30yr	30/360	RE	HU	M	M	14
Skinner §42 Ex 1	$1k / 6% / 15yr	30/360	RE	HU	A	A	–
Skinner §42 Ex 3 piano	$500 / 6% / 5yr	30/360	RE	HU	A	M	–
Synthetic HALF_EVEN	$100k / 4.80% / 30yr	30/360	RE	HU	M	M	–
Synthetic HALF_UP	$100k / 4.80% / 30yr	30/360	RE	HU	M	M	–
Synthetic ROUND_UP	$100k / 4.80% / 30yr	30/360	RE	UP	M	M	–
Wikipedia mortgage calc	$200k / 6.5% / 30yr	30/360	RE	UP	M	M	–
0% Promo	$999 / 0% / 2yr	30/360	RE	HU	M	M	–

41 fixtures total.

Column codes. Day: `30/360` US residential, `A/360` Actual/360 US commercial. Bal: `RE` round-each-balance, `CP` carry-precision. Rnd: `UP` `ROUND_UP`, `HU` `ROUND_HALF_UP`, `HE` `ROUND_HALF_EVEN`. Cmp: `M` monthly, j_2 semi-annual (Canadian *Interest Act* §6), `A` annual. Freq: `M` `SM` `BW` `W` `Q` `A` (monthly to annual). ARM: count of `RateChange` entries; `cap` indicates at least one `payment_cap_factor`. A dash is fully fixed-rate.

Bibliography

Arcones SOA FM §4.1 Ex 4. Arcones, M. A. *Manual for SOA Exam FM / CAS Exam 2*, Chapter 4 "Amortization and sinking bonds," §4.1 Example 4. 2009. https://people.math.binghamton.edu/arcones/exam-fm/sect-4-1.pdf

Broverman MIC §2.1 Ex 2.7. Broverman, S. A. *Mathematics of Investment and Credit*, 8th ed., §2.1 Example 2.7(a)–(b). ACTEX, 2023. https://www.actexlearning.com/samples/MIC_8th_edition_051923_SAMPLE.pdf

CFPB H-25(B). Consumer Financial Protection Bureau. *Closing Disclosure (Sample), Form H-25(B).* 2014. https://files.consumerfinance.gov/f/201403_cfpb_closing-disclosure_cover-H25B.pdf

CFPB Interest-Only Loan. CFPB sample Loan Estimate for an Interest-Only loan. https://www.consumerfinance.gov/policy-compliance/guidance/mortgage-resources/tila-respa-integrated-disclosures/forms-samples/

eCampus §4.3 (Pearline / Erika / Johnetta). eCampus Ontario. *Mathematics of Finance*, §4.3 "Loan amortization: formula approach" (Example 4.3.1 Pearline; Exercises 2 Erika and 3 Johnetta). Pressbooks (CC-BY-NC-SA 4.0), 2023. https://ecampusontario.pressbooks.pub/financemath/chapter/4-3-loan-amortization-formula-approach/

eCampus §4.4.1 (T1 + renewal). eCampus Ontario. *Mathematics of Finance*, §4.4.1 "Mortgages: Formula approach." Pressbooks (CC-BY 4.0). https://ecampusontario.pressbooks.pub/financemath/chapter/4-4-mortgages-formula-approach/

FNMA §1103. Federal National Mortgage Association. *Multifamily Selling and Servicing Guide*, §1103 "Actual/360 Tier 2 SARM Loan Amortization Calculation." Effective 2026-04-03. https://mfguide.fanniemae.com/node/5286

FHLBB *Review* (March 1935). Federal Home Loan Bank Board. "Direct-Reduction Plan vs. Share-Accumulation Plan," *Federal Home Loan Bank Review*, Vol. 1, No. 6 (March 1935), pp. 187–198. FRASER: https://fraser.stlouisfed.org/title/federal-home-loan-bank-review-116/march-1935-2083

Geltner Ch 20. Geltner, D., Miller, N. G., van de Minne, A., Eichholtz, P., Lindenthal, T., & Shen, J. *Commercial Real Estate Analysis and Investments*, online supplement Chapter 20: "Mortgage Basics II — Payments, Yields, and Values," Exhibit 20-6 (Constant-Payment Mortgage). Routledge, 2024. ISBN 9781041081197.

Goldstein §10.3. Goldstein, L. J., Schneider, D. I., Lay, D. C., & Asmar, N. H. *Finite Mathematics and Its Applications*, 12th ed., §10.3 Example 1 + Table 1. Pearson, 2018. ISBN 978-0-13-443776-7. https://www.pearsonhighered.com/assets/samplechapter/0/1/3/4/0134437764.pdf

JHF Flat 35. Japan Housing Finance Agency (☒☒☒☒☒☒☒☒) Flat 35 mortgage repayment comparison page. ¥20,000,000 at 1.5% fixed over 30 years, ganri kinto hensai (☒☒☒☒☒☒, equal total payment / standard annuity). https://www.flat35.com/hajimete/atoz/04.html

Las Positas §8.05. Las Positas College Mathematics. *Math for Liberal Arts*, §8.05 "Amortized Loans," Examples 1 and 3. LibreTexts (CC-BY). https://math.libretexts.org/Courses/Las_Positas_College/Math_for_Liberal_Arts/08:_Consumer_Mathematics/8.05:_Amortized_Loans

MoneyVox Tableau d'amortissement. MoneyVox worked example of a tableau d'amortissement (French amortization schedule). €10,000 at 5% over 12 monthly payments with 0.35% assurance emprunteur. https://www.moneyvox.fr/credit/tableau-amortissement.php

MS State P3920. Mississippi State University Extension Service. Publication P3920, "Time Value of Money for Beginning Farmers and Ranchers." https://extension.msstate.edu/sites/default/files/publications/P3920_web.pdf

Olivier Chans (T1 + renewal). Olivier, J. *Business Math: A Step-by-Step Handbook*, §13.4 "Special Application: Mortgages" (the Chans first term and renewal). LibreTexts (CC-BY-NC-SA), 2021. https://math.libretexts.org/Bookshelves/Applied_Mathematics/Business_Math_(Olivier)/13:_Understanding_Amortization_and_its_Applications/13.04:_Special_Application_-_Mortgages

OpenStax (Contemporary Mathematics). OpenStax College. *Contemporary Mathematics*, §6.8 "The Basics of Loans," §6.12 "Renting and Homeownership," and Chapter 6 Answer Key (problems 6.36, 6.78, 6.100, 6.110, 6.114). CC-BY-4.0. https://openstax.org/books/contemporary-mathematics

ProEducate ARM cap. ProEducate. *ARM Payment Caps* (worked example derived from a Federal Reserve consumer-handbook lineage, mirrored at mortgagesfinancingandcredit.org). 2014. https://www.proeducate.com/courses/Finance/PaymentCap.pdf

RBC Accelerated Bi-Weekly. RBC Accelerated Bi-Weekly mortgage worked example. https://www.rbcroyalbank.com/mortgages/accelerated-bi-weekly-payments.html

Reg Z H-14. U.S. Code of Federal Regulations. 12 CFR Part 1026 (Regulation Z, Truth in Lending), Appendix H, Sample H-14: *Variable-Rate Mortgage Sample*. https://www.ecfr.gov/current/title-12/chapter-X/part-1026/appendix-Appendix%20H%20to%20Part%201026

Skinner (1913). Skinner, E. B. *The Mathematical Theory of Investment*, §42 Examples 1 and 3. Boston: Ginn and Company, 1913. Public domain. https://archive.org/details/mathematicaltheo00skin

Synthetic HALF_UP / ROUND_UP / HALF_EVEN. Synthetic boundary case constructed for this library: $100,001.25 / 4.8% / 30yr is engineered so month-1 unrounded interest equals exactly $400.005 — a half-cent boundary that distinguishes `ROUND_HALF_UP` ($400.01) from `ROUND_HALF_EVEN` ($400.00) and from `ROUND_UP` (which rounds the payment up to $524.68 instead of $524.67).

Wikipedia mortgage calc. Wikipedia contributors. "Mortgage calculator." *Wikipedia, The Free Encyclopedia*. Worked example in the monthly-payment formula section. https://en.wikipedia.org/wiki/Mortgage_calculator

Zero-Interest Promotional Financing. Synthetic fixture modeling 0% APR promotional financing (e.g. consumer electronics or furniture loans).